

Terms. The model then predicts relations between nodes belonging to seven relation types: *Used-for*, *Feature-of*, *Evaluate-for*, *Hyponym-of*, *Part-of*, *Compare*, and *Conjunction*. Because we want to generate new ideas, we focus on *used-for* relations in papers. Next, we use SciCo (Cattan et al., 2021) with checkpoint from Hugging Face¹² to obtain entity coreference to merge identical nodes. Then, we use ScispaCy (Neumann et al., 2019) to perform unsupervised abbreviation detection to replace the abbreviation with a more informative long form. Finally, we perform scientific sentence classification (Cohan et al., 2019)¹³ to classify sentences from the abstract into five categories including *Background*, *Method*, *Objective*, *Other*, and *Result*. We select sentences with labels of *Background* and *Other* as background context. During preprocessing, we only keep high-confidence outputs from IE models. Figure 4 shows an example of the IE systems pipeline.

A.2 Biochemical Dataset Collection

We collect PubMed papers from 1988 to 2024 using Entrez Programming Utilities API¹⁴ for the following topics, including *Yarrowia*, *Saccharomyces cerevisiae*, *Issatchenkia orientalis*, and *Rhodospiridium toruloides*. We use PubTator 3 (Islamaj et al., 2021; Wei et al., 2022; Luo et al., 2023; Wei et al., 2023; Lai et al., 2023). The PubTator 3 performs named entity recognition, relation extraction, entity coreference and linking, and entity normalization for the abstracts in the dataset. PubTator 3 identifies bio entities belonging to seven types: *gene*, *chemical*, *chromosome*, *cell line*, *variant*, *disease*, and *species* and relations belonging to 13 types: *associate*, *cause*, *compare*, *convert*, *contract*, *drug interact*, *inhibit*, *interact*, *negative correlate*, *positive correlate*, *prevent*, *stimulate*, and *treat*. Finally, we use a sentence classifier trained on CODA-19 (Huang et al., 2020) to classify sentences in abstracts into *background*, *purpose*, *method*, *finding*, and *other*. We select sentences with labels of *background* as background context and remove sentences with labels of *other*. We treat the rest sentences that have at least one entity as the target sentence. We only keep samples with low similarity between background context

and corresponding ground truth sentences.¹⁵ Our final dataset has 4,767 papers before 2023/02, 642 papers from 2023/02 to 2023/08, and 299 papers after 2023/08.

B Finetuning and Automated Evaluation details

B.1 Inspiration Retrieval Module

The statistics of each inspiration type are in Table 7. Table 8 shows sample retrieved inspirations.

B.1.1 Semantic Neighbors

We use all-mpnet-base-v2 from SentenceBert (Reimers and Gurevych, 2019), which performs best in semantic search to retrieve similar nodes from the training set based on query q in §3.1. We retrieve up to 20 relevant semantic neighbors $\mathcal{R}$ from the training set for each instance. We treat the target nodes from $\mathcal{R}$ as semantic neighbors.

B.1.2 KG Neighbors

We use one-hop connected neighbors from the background KG $\mathcal{G}_B$ constructed on papers before 2021 (i.e., the papers in the training set). Because of the scarcity of KG neighbors, we do not limit the number of KG neighbors.

B.1.3 Citation Neighbors

Similar to semantic neighbors, we use all-mpnet-base-v2 from SentenceBert (Reimers and Gurevych, 2019) to retrieve cited paper titles similar to query q . We restrict cited papers only before 2021. We retrieve up to 5 relevant citation neighbors from the papers' citation network.

B.2 Generation Module

Our T5 model and their variants are built based on the Huggingface framework (Wolf et al., 2020).¹⁶ We optimize those models by AdamW (Loshchilov and Hutter, 2019) with the linear warmup scheduler.¹⁷ Those models are finetuned on 4 NVIDIA A6000 48GB GPUs with distributed data parallel.¹⁸ The training time for each model is about 10 hours.

¹⁵The similarity is calculated with all-mpnet-base-v2.

¹⁶github.com/huggingface/transformers

¹⁷huggingface.co/docs/transformers/main_classes/optimizer_schedules#transformers.get_linear_schedule_with_warmup

¹⁸pytorch.org/tutorials/intermediate/ddp_tutorial.html

¹²huggingface.co/allenai/longformer-scico

¹³github.com/allenai/sequential_sentence_classification

¹⁴www.ncbi.nlm.nih.gov/books/NBK25501/

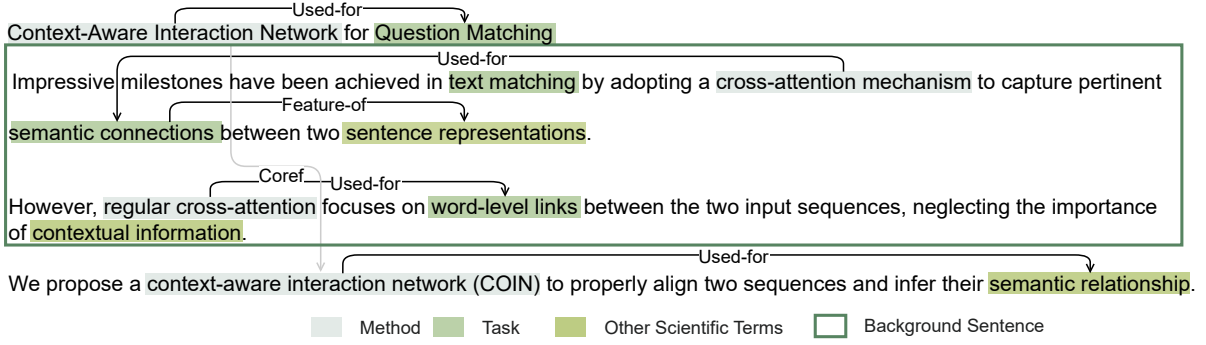


Figure 4: Preprocessing result for Hu et al. (2021) in non-canonicalized KG Corpus

Stage	PL-Maker Entities	PL-Maker Used-for Relations	SciCo Coreference	Scispacy Abbreviation Detection	Sentence Classification
Precision	91.3	65.4	97.2	100	100

Table 6: Human quality evaluation of preprocessing stages(%). Overall pass rate after all steps are applied is 79.7%.

Type	Train	Valid	Test
SN	10.8	10.0	10.0
KG	8.3	8.0	8.1
CT	4.9	5.0	5.0

Table 7: Average of # of neighbors for each instance, excluding those which do not have any neighbor

B.2.1 In-Context Learning

We choose GPT3.5 davinci-003¹⁹ (Brown et al., 2020) as our out-of-the-box causal language modeling baseline. We select 5 instances from the training set as examples for the few-shot setting. We randomly select those examples for GPT3.5FS. For GPT3.5Retr, similar to semantic neighbors, we use all-mpnet-base-v2 from SentenceBert (Reimers and Gurevych, 2019), which performs best in semantic search to retrieve similar instances from the training set based on query q in §3.1. The input length is limited to 2048 tokens due to OpenAI API limits. We choose gpt-4-0314 as our GPT4 model. Our input for GPT4 is similar to GPT3.5.

For each selected example from the training set with *forward* relation, the template is “Consider the following context: $\mathcal{M}$ In that context, which p can be used for v , and why? $\mathcal{T}$ ”, where $\mathcal{M}$ is the background context, p is the target node type, v is the seed term, and $\mathcal{T}$ is the target idea sentence; for *backward* relation, the template is “Consider the following context: $\mathcal{M}$ In that context, which p do we use v , and why? s ”. For selected examples with

additional retrieval inspirations, we concatenate the following additional template to the $\mathcal{M}$: “The retrieval results are: $i_1, \dots, i_k$ ”, where $i_1, \dots, i_k$ are retrieved inspirations. For the final prompt, the template is similar to the above example template. However, the target sentence $\mathcal{T}$ will not be included. We ask the model to generate 10 outputs. We will select the best output and skip the empty output.

B.2.2 Fine Tuning

Given input without any inspirations, the input combines the prompt $\mathcal{P}$ and context $\mathcal{M}$ as shown in §3.1 (i.e., $\mathcal{P} \mid \text{context: } \mathcal{M}$). Given input with inspirations, the input is $\mathcal{P} \mid \text{retrieve: } i_1, \dots, i_k \mid \text{context: } \mathcal{M}$, with $i_1, \dots, i_k$ as retrieved inspirations. The input length is limited to 512 tokens. For both tasks, we finetune our model based on T5-large with a learning rate of 6×10^{-6} and $\epsilon = 1 \times 10^{-6}$. The batch size is 8 for each GPU. The maximum training epoch for all models is 10 with 4 patience. During decoding, we use beam-search to generate results with a beam size of 5 and a repetition penalty of 1.5.

In-context Contrastive Augmentation We randomly select 2 sentences that appeared in the input as in-context negatives. For example, in Figure 1, the in-context negatives could be “knowledge acquisition is done by using Method”, “this requires plms to integrate the information from all the sources in a lifelong manner.”.

¹⁹openai.com/api/

B.2.3 Biochemical Case Study

Our Meditron-7b (Chen et al., 2023) and its variants are built based on the Huggingface framework (Wolf et al., 2020).²⁰ We use its `epfl-llm/meditron-7b` as the base model. We finetune those models with a learning rate of 2×10^{-6} and $\epsilon = 5 \times 10^{-8}$. The maximum training epoch for all models is 5. All models are finetuned on 4 NVIDIA A100 80 GB GPUs with Fully Sharded Data Parallel.²¹ The training time for each model is about 20 hours.

B.3 The Scale of Retrieval Set

We retrieve from a set of 59k papers with over 374k sentences in the NLP domain, the focus of our experiments. Our background KG built on the training set has more than 197k nodes and 261k relations. Moreover, we collect 87k paper titles from citation networks. This represents a large-scale and diverse domain; retrieving inspirations from this set is expected, in principle, to be more than enough for generating novel ideas. Indeed, NLP papers typically cite each other and build on each other as inspirations to create new ideas - which motivates our inspiration retrieval.

B.4 Automated Evaluation

We use BERTScore (Zhang* et al., 2020) with SciBERT checkpoint for both tasks. The hash of the checkpoint is `allenai/scibert_scivocab_uncased_L8_no-idf_version=0.3.12(hug_trans=4.19.2)`. The automated evaluation results are in Table 9.

C Human Annotation and Evaluation Details

Gold Dataset Annotation Details The gold dataset annotation interface is in Figure 5. The quality of the instances in the test set is judged given three criteria: (1) whether the ground truth sentence trivially overlaps with background context; (2) whether background context contains relevant information for the target relation; (3) whether the target relation (from which the seed term is taken) is a salient aspect of the idea proposed in the target paper.

Study I The instructions for human evaluation can be found in Figure 6, while an example of the

human evaluation interface is provided in Figure 7 and 8. Human annotators are required to evaluate each system output based on the following criteria: (1) *Is the candidate relevant to the context + seed term?* (2) *Does the candidate copy too much from the context, or is it sufficiently novel/different from the context?* (3) *Does the candidate’s suggestion generally make sense to you scientifically?* (4) *Is the language sufficiently clear and coherent to understand the suggestion?* The input for sample human annotation is in Table 10 and the human labels are in Table 11. The human annotation agreement is in Table 13.

Study III We ask the following questions to human annotators to evaluate the quality of regeneration results: (1) *Is the regenerated idea substantially different from the original?* (2) *Is the regenerated idea more novel and creative than the original idea?* (3) *Does the second iteration increase novelty?* The human annotation agreement is in Table 14.

D Scientific Artifacts

We list the licenses of the scientific artifacts used in this paper: Semantic Scholar Academic Graph API (API license agreement²²), Huggingface Transformers (Apache License 2.0), SBERT (Apache-2.0 license), BERTScore (MIT license), Meditron-7b (Llama2), Entrez Programming Utilities API (Copyright²³), PubTator 3 (Data use policy²⁴), and OpenAI (Terms of use²⁵).

E Ethical Consideration

The SCIMON task and corresponding models we have designed in this paper are limited to the natural language processing (NLP) and biochemical domain, and might not apply to other scenarios.

E.1 Usage Requirement

This paper aims to provide investigative leads for a scientific domain, specifically natural language processing. The final results are not intended to be used without human review. Accordingly, domain experts might use this tool as a research writing assistant to develop ideas. However, our system does not do any fact-checking with external knowledge. In addition, we train our models on the ACL

²⁰github.com/huggingface/transformers

²¹https://huggingface.co/docs/accelerate/usage_guides/fsdp

²²api.semanticscholar.org/license/

²³www.ncbi.nlm.nih.gov/books/about/copyright/

²⁴www.ncbi.nlm.nih.gov/home/about/policies/

²⁵openai.com/policies/terms-of-use